

Prediction and Monitoring of Vessel Speed under Environmental Influences Using Machine Learning

Name: Muhammad Hafiz Hamzah bin Fansuri

Supervisor: Assoc. Prof. Ir. Dr. Noor Hazrin Hany

List of Contents

01. Introduction

02. Problem statement

03. Objective

04. Project Scope

05. Methodology

06. Grey box model

07. System Architecture

08 Hardware Setup

09. Iterative road testing

10. Result

11. Key outcomes & findings

12. Conclusion

Introduction

The shipping industry is the backbone of global trade, moving goods across oceans every day. A key factor in ship operations is speed, which influences fuel use, costs, and timely deliveries. Environmental conditions like wind, waves, and currents can affect how vessels move, making it important to understand their impact. Modern sensors, GPS, and tracking systems allow us to collect vast amounts of data, and machine learning can help uncover patterns that improve predictions of vessel speed and enable real-time monitoring, providing operators with useful insights to maintain efficiency and safety.

Problem statement

1. Incomplete integration of environmental and design factors in predictive models.
2. The lack of sufficient hybridization between physics-based and data-driven methods to ensure both accuracy and physical interpretability.
3. Limited validation across different real-world conditions restricts generalization for various global maritime applications.

OBJECTIVES

1. To develop a machine learning model that can accurately predict vessel speed under different environmental conditions.
2. To implement a vessel speed compliant prototype with a real-time monitoring dashboard that displays vessel speed, forecasts, and threshold-based alerts.
3. To validate the functionality and performance of the machine learning and monitoring system.

Project scope

1. Wind speed & wave height effects on vessel speed
2. PINN + XGBoost grey-box prediction model
3. ESP32 hardware prototype with real sensors
4. Road testing at IIUM campus
5. Real-time Blynk dashboard with speed alerts

literature review & Findings

Model	Capabilities	Limitation	Citation
Grey-Box Model (GBM: PINN + XGBoost)	Integrates physics and data, robust with limited data, high accuracy	Needs physics-based model, more complex to implement	Lang et al., 2024
Generalized Regression Neural Network (GRNN)	Handles nonlinearities, robust to noise, best among tested ML models	Requires large, quality data; less interpretable	Yuan et al., 2021
Artificial Neural Network (ANN)	High accuracy, flexible, robust to missing data	Needs large datasets, risk of overfitting	-Bassam et al., n.d. -Tarelko & Rudzki, 2020 - Zhou et al., 2022
XGBoost (Extreme Gradient Boosting)	Handles complex relationships, good with tabular data	Less interpretable, needs tuning	-Lang et al., 2022 - Lang & Mao, 2020
Random Forest Regression (RFR)	Handles nonlinearities, robust to outliers	Computationally intensive, less interpretable	-Abebe et al., 2020 - Zhou et al., 2022 - Lang & Mao, 2020
Gaussian Process Regression (GPR)	Provides uncertainty estimates, good with small data	Computationally expensive, less scalable	- Abebe et al., 2020) -Bassam et al., n.d.
Support Vector Regression (SVR)	Good with high-dimensional data, robust	Sensitive to parameter selection, less scalable	- Moreira et al., 2021 - Zhou et al., 2022 - Lang et al., 2022
Linear Regression	Simple, interpretable	Poor with nonlinearities, lower accuracy	-Bassam et al., 2022 -El Mekkaoui et al., 2023
LSTM (Long Short-Term Memory)	Captures temporal dependencies, good for time series	Needs large sequential data, complex tuning	- Wang et al., 2022 -El Mekkaoui et al., 2023
Semi-Empirical/Process-Based	Physically interpretable, validated with experiments	Limited to specific conditions, less flexible	-Tarovik et al., 2024 - Lang & Mao, 2020

- Limited generalisation across vessel types and sea conditions
- Simplified or incomplete modelling of environmental factors
- ML models lack physical interpretability
- Few studies integrate multiple environmental variables
- Small or region-specific datasets limit practical use

Methodology



Phase 1: Historical Data Analysis

- Analyzed 4 months of operational data from vessel Jati 6 (April–July 2025).

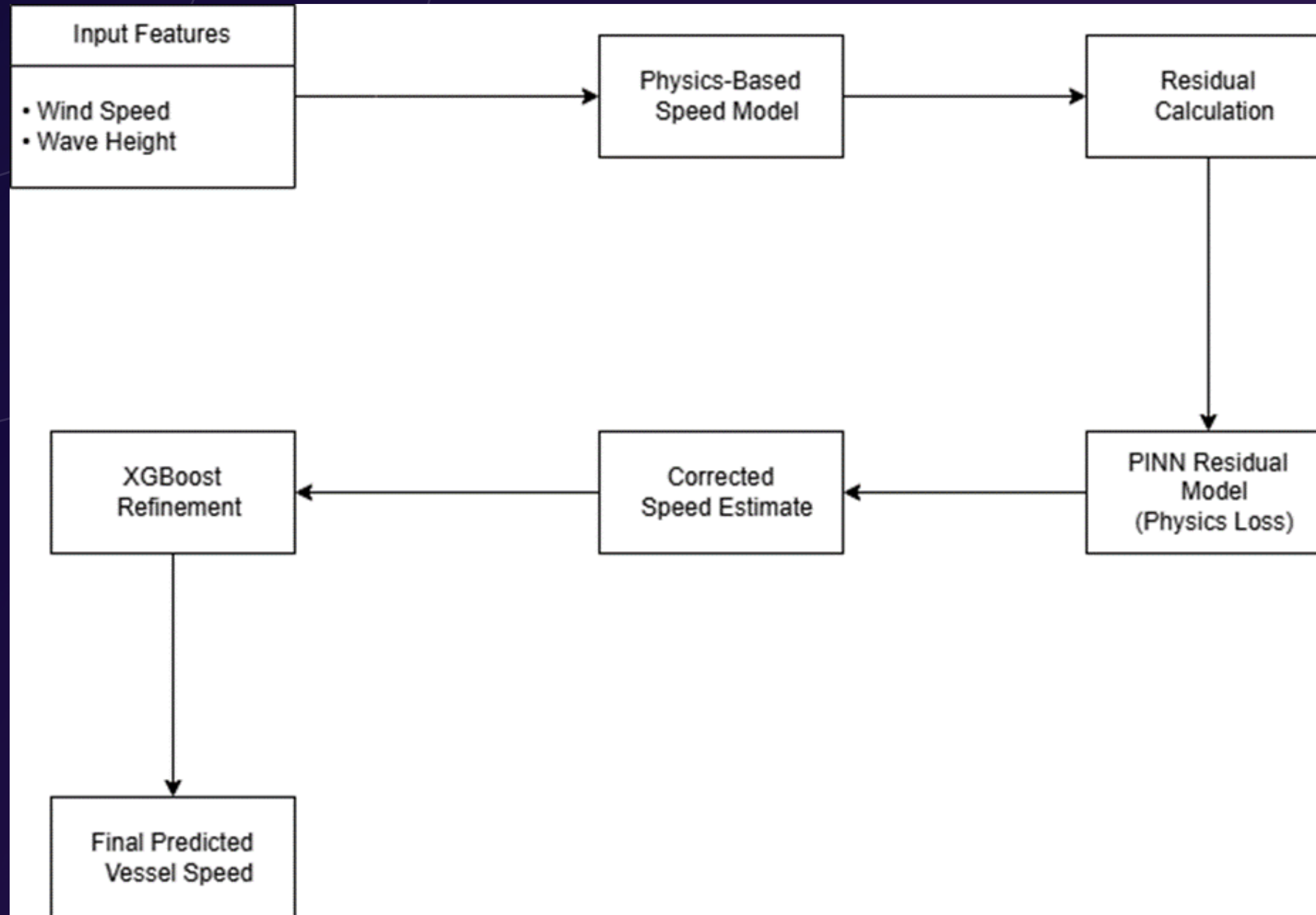
Phase 2: Hybrid Grey-Box Model Development

- Combined Physics-Informed Neural Networks (PINN) with XGBoost.
- Integrated physical laws of ship resistance with data-driven accuracy.

Phase 3: Hardware Integration

- ESP32 Microcontroller: Central processing unit.
 - Airmar 200WX Weather Station: Measured wind speed and GPS SOG.
 - BNO085 IMU: Measured wave height via linear acceleration.
- 

Grey-Box Prediction Model



Combines physics-based reasoning and machine learning for accurate, interpretable predictions.

Physics-informed component:

- Captures effects of wind and waves
- Ensures physically realistic predictions using simplified relationships.

Machine learning component (XGBoost):

- Models nonlinear interactions from sensor data.
- Inputs: wind, waves, SOG and derived features.

Law and Equation

Total Resistance Principle

$$\mathbf{R}_T = \mathbf{R}_0 + \Delta \mathbf{R}_{AW} + \Delta \mathbf{R}_{AA}$$

Physics Baseline Prior

$$V_{phys}(W, H) = \beta_0 + \beta_1 W + \beta_2 H + \beta_3 W^2 + \beta_4 H^2 + \beta_5 WH$$

PINN Monotonicity Loss

$$L = \text{MSE}(f_{pinn}(x), r) + \lambda [\text{ReLU}(\Delta f_{pinn, \text{wind}}) + \text{ReLU}(\Delta f_{pinn, \text{wave}})]$$

Training Configuration

Dataset Splitting Ratio:

- 57% Train | 10% Validation | 15% Test

Stage 1:

- Physics Prior (Ridge Regression) Regularisation (α): 10.0

Stage 2: Residual Learning (PINN)

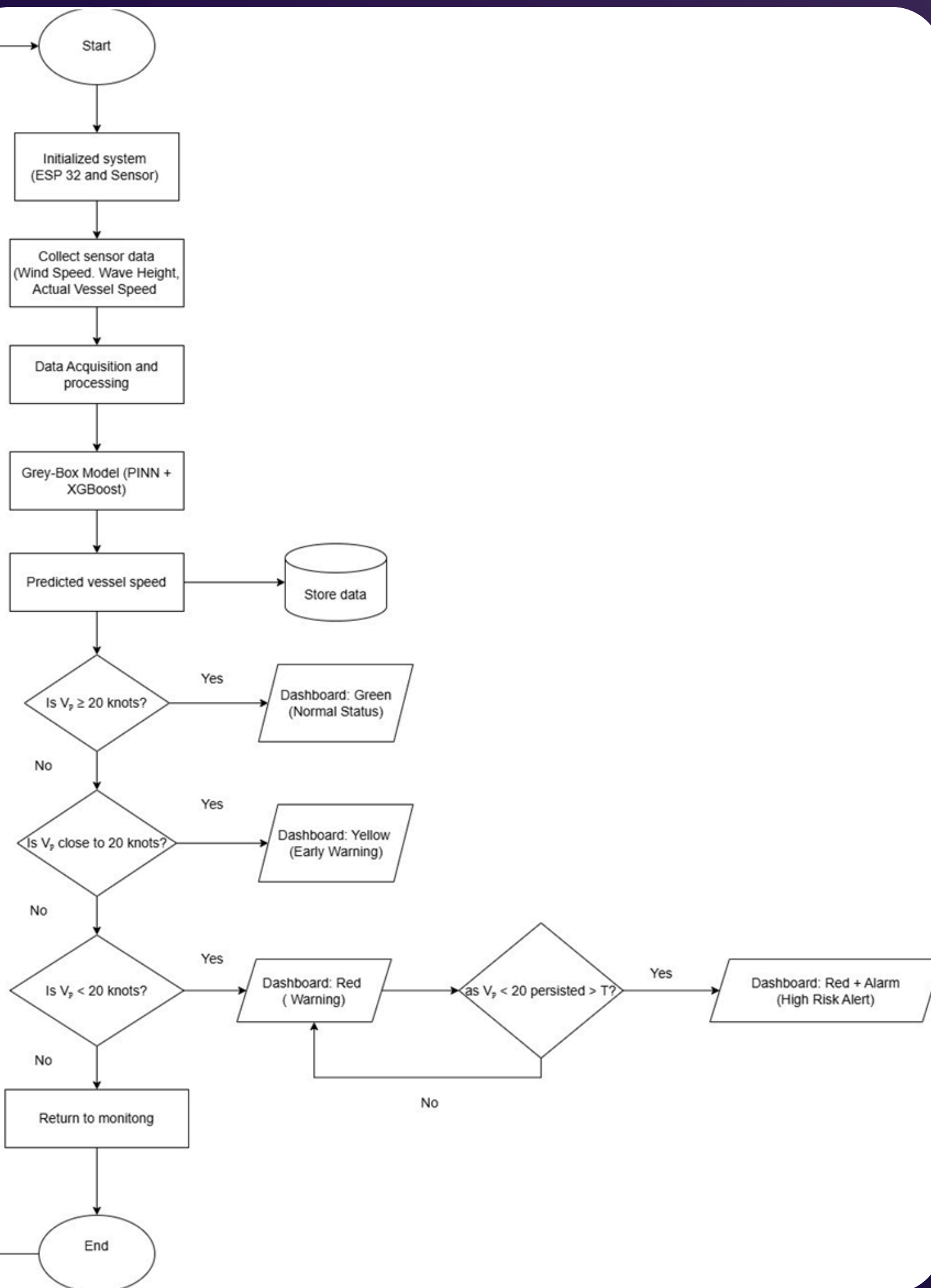
- Architecture: 4 Residual Blocks | 128 Hidden Dimensions
- Optimisation: Learning Rate = 1×10^{-3} | Max Epochs: 500 (Patience: 80)
- Penalty Weight λ : 0.5

Stage 3: Residual Correction (XGBoost)

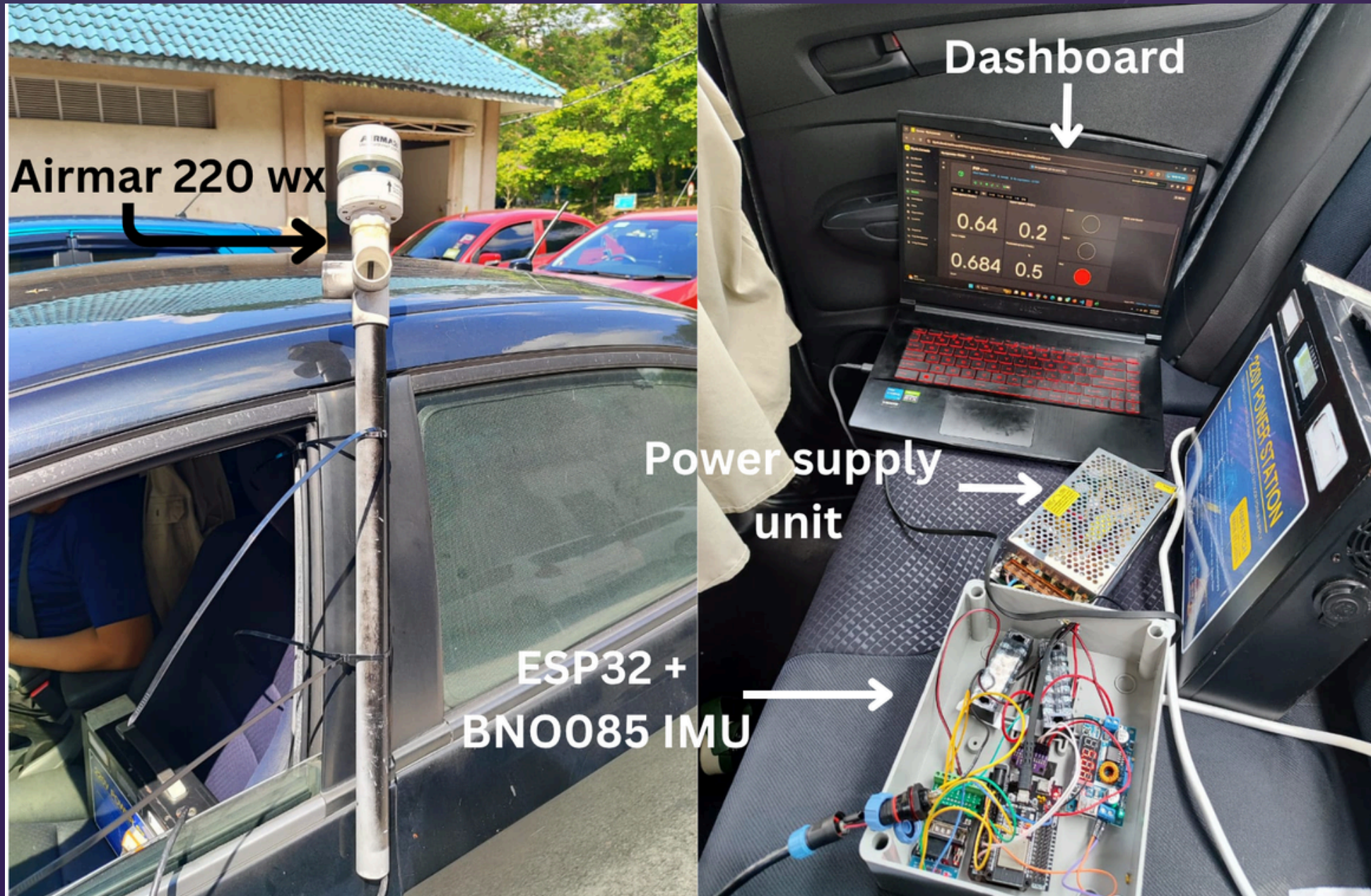
- Parameters: 1000 Estimators | Max Depth: 5 | Learning Rate: 0.02
- Regularisation: L1 = 0.1 | L2 = 2.0
- Early Stopping: 50 Rounds

System Architecture

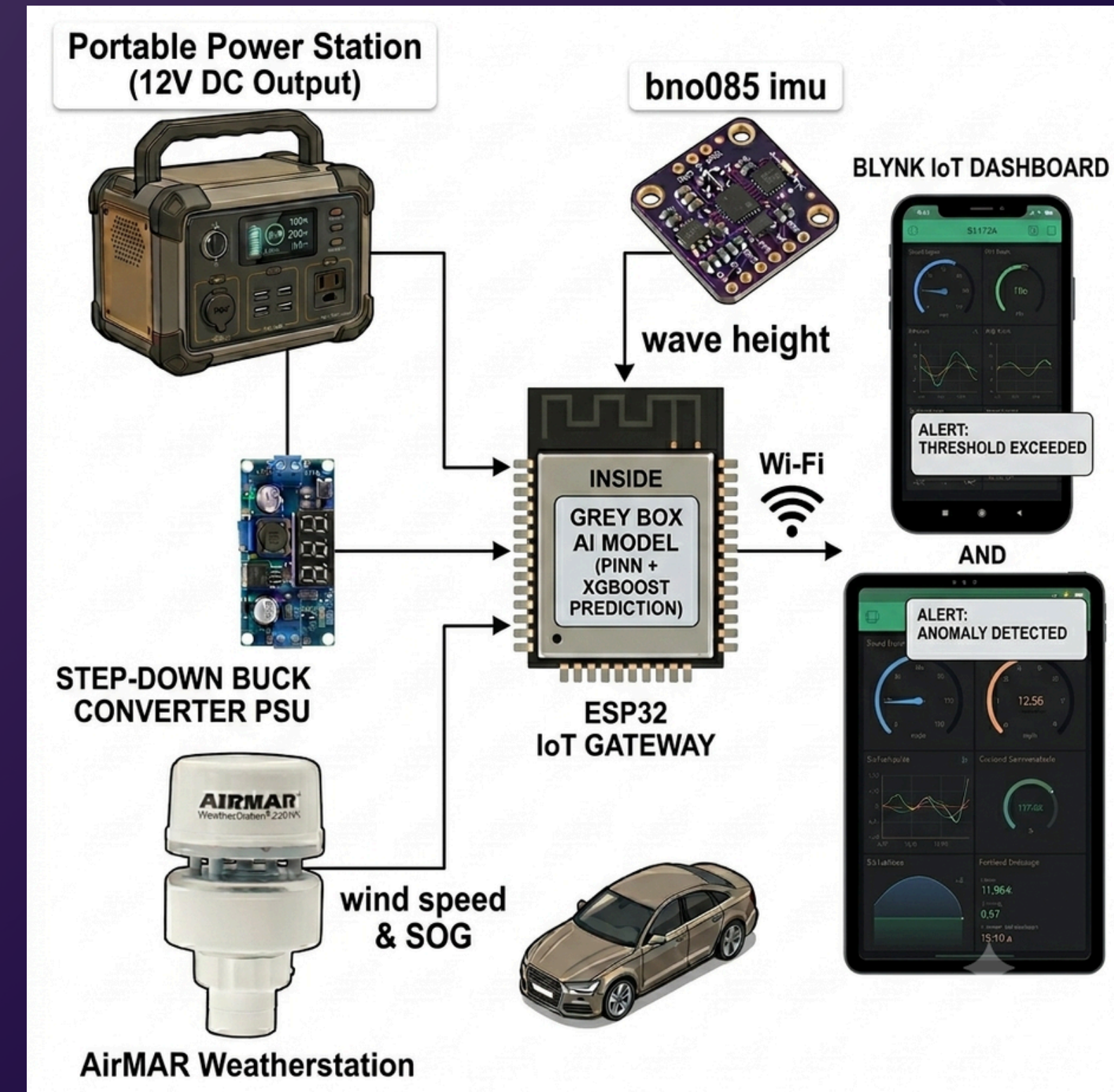
- Sensors → Grey-Box Model → Real-Time Alert
- ESP32 collects wind speed, wave height & vessel speed.
- PINN + XGBoost predicts speed.
- Dashboard alerts GREEN / YELLOW / RED.



Hardware Setup



- Airmar 220 wx : wind speed & SOG
- Bno085 imu : wave height

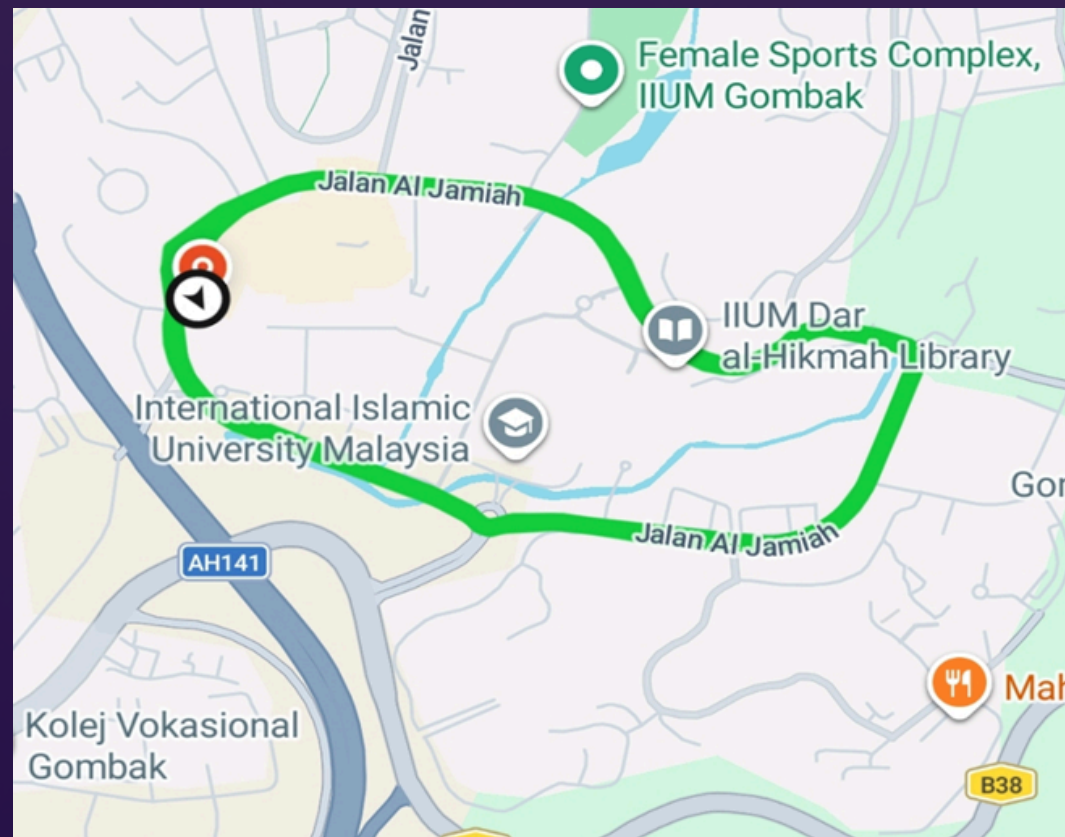


ITERATIVE ROAD TESTING

Test 1 (3 May)
Maritime model
Baseline check

Test 2 (30 May)
Road- retrained
Domain fix

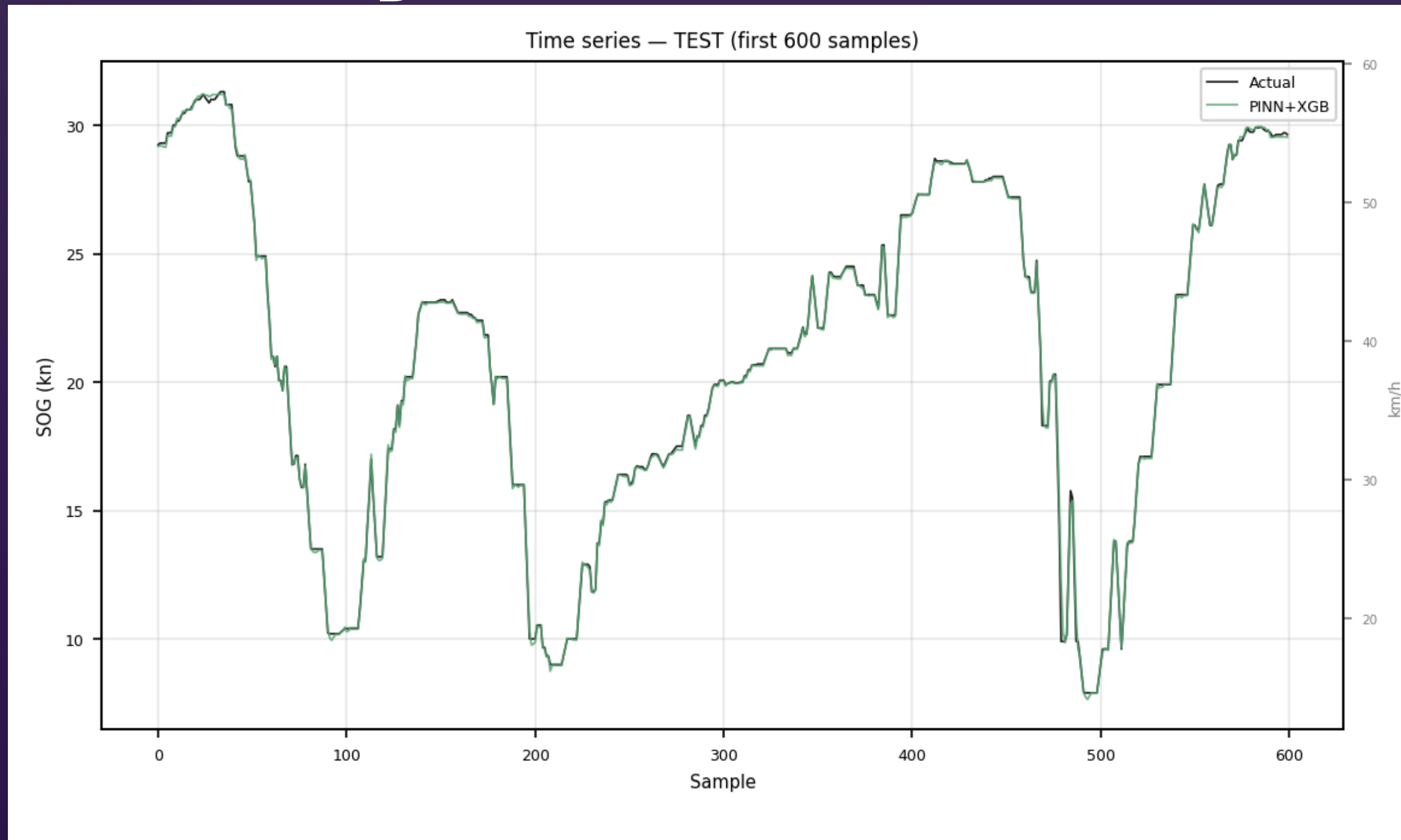
Test 3 (6 Jun)
Combined dataset
Final model



Route features

- ~2.7 km per round
- 12 speed humps per round

Grey box results



Final model trained on combined road dataset
(Test 1 + Test 2 data)

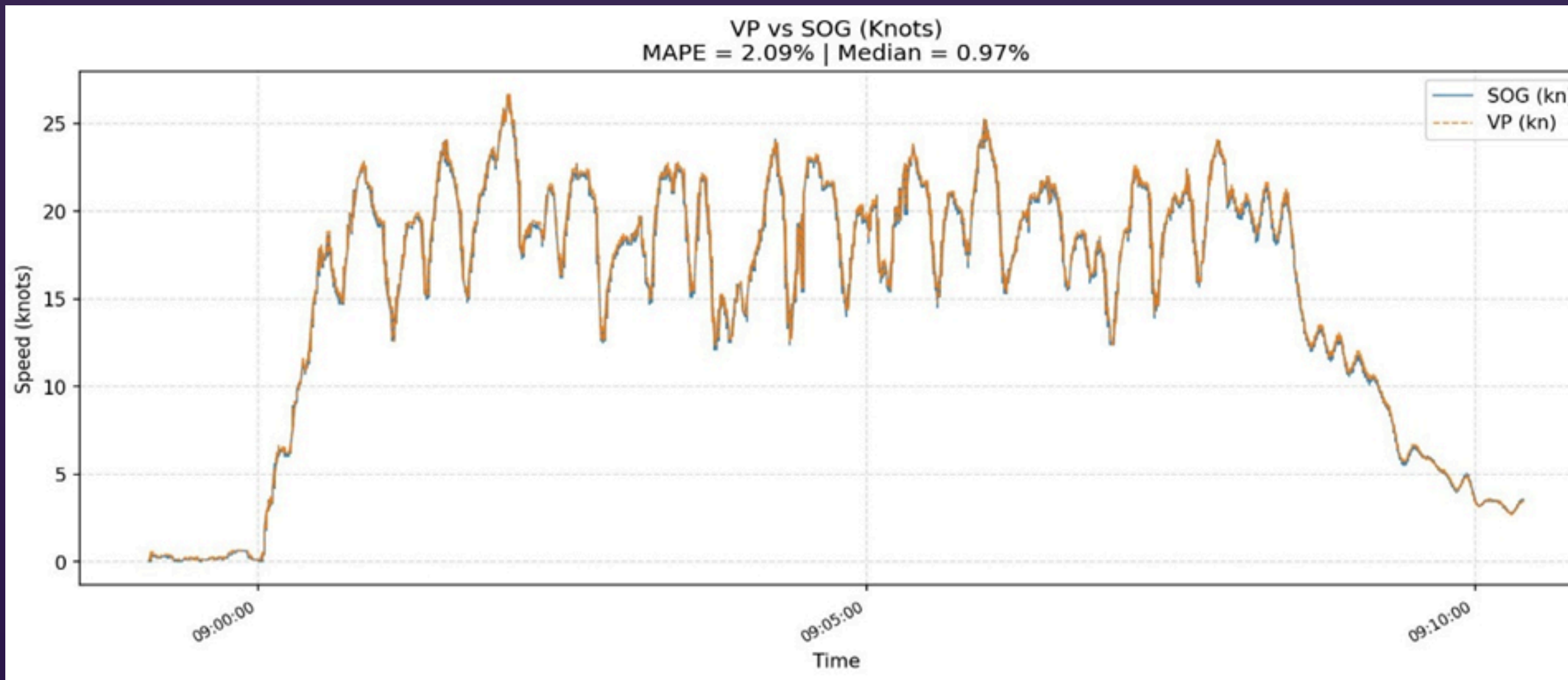
Results

RMSE
0.074 knots

R^2
0.9998

WITHIN ± 0.5 kn
99.6 %

Testing results



Training

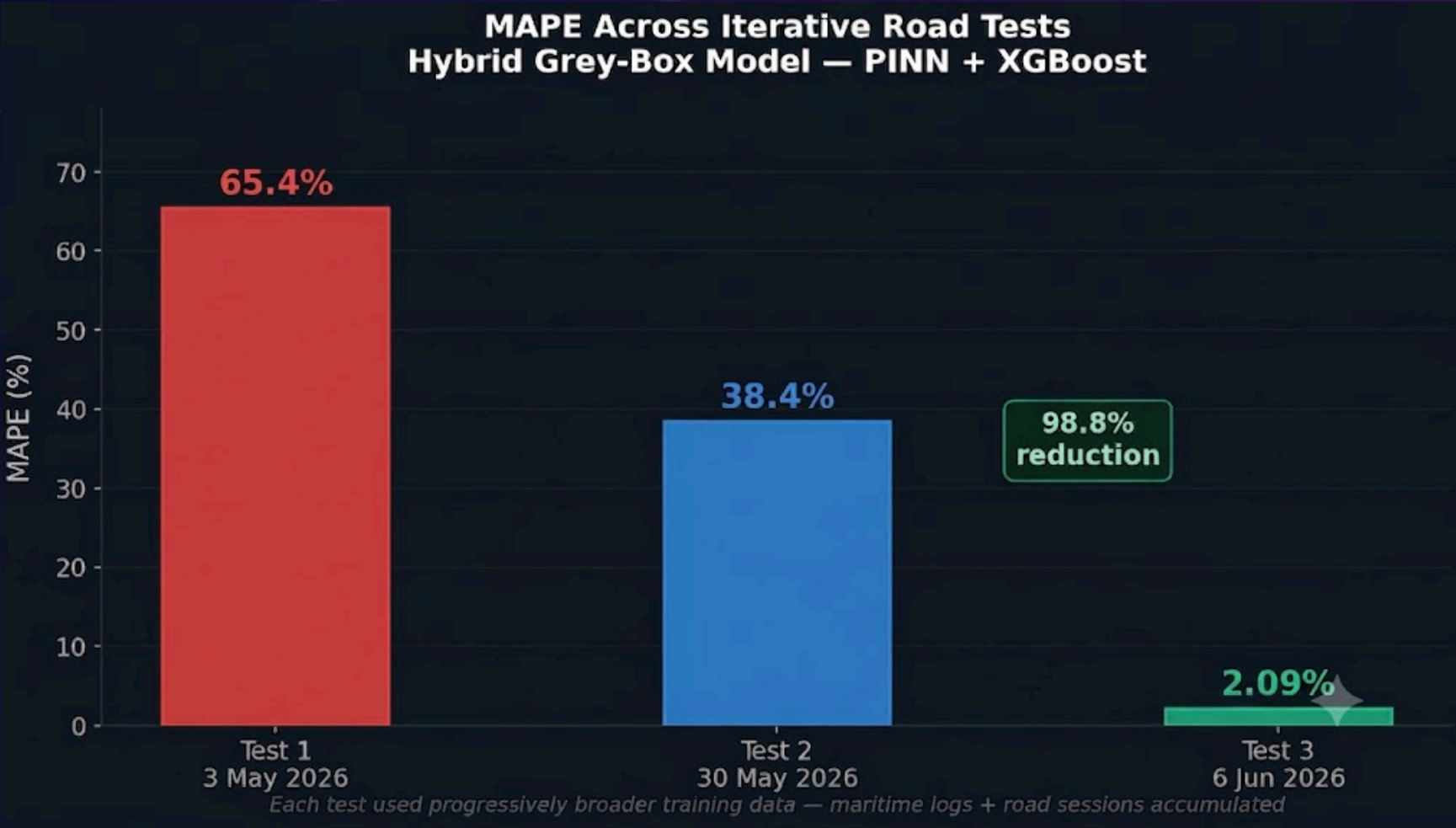
- Combined Test 1 + Test 2 data

Model Performance

- RMSE: 0.187
- R^2 : 0.9989
- MAPE: 2.09%
- 99.8% of predictions within ± 0.5 knots

Test 1 $\rightarrow$ 15.05 knots RMSE $\rightarrow$ Test 2 $\rightarrow$ 6.59 knots $\rightarrow$ Test 3 $\rightarrow$ 0.187 knots

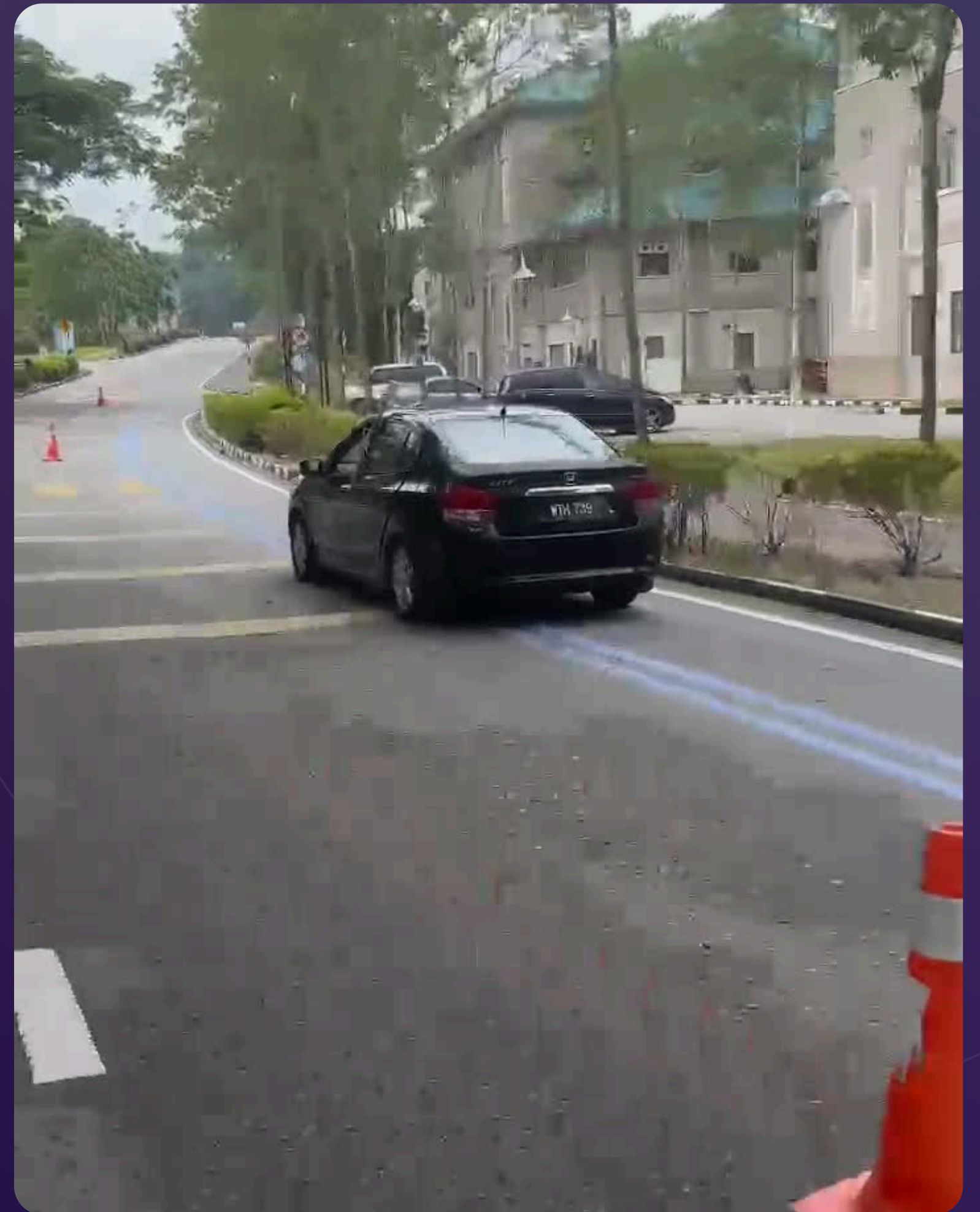
Key Outcomes & Findings



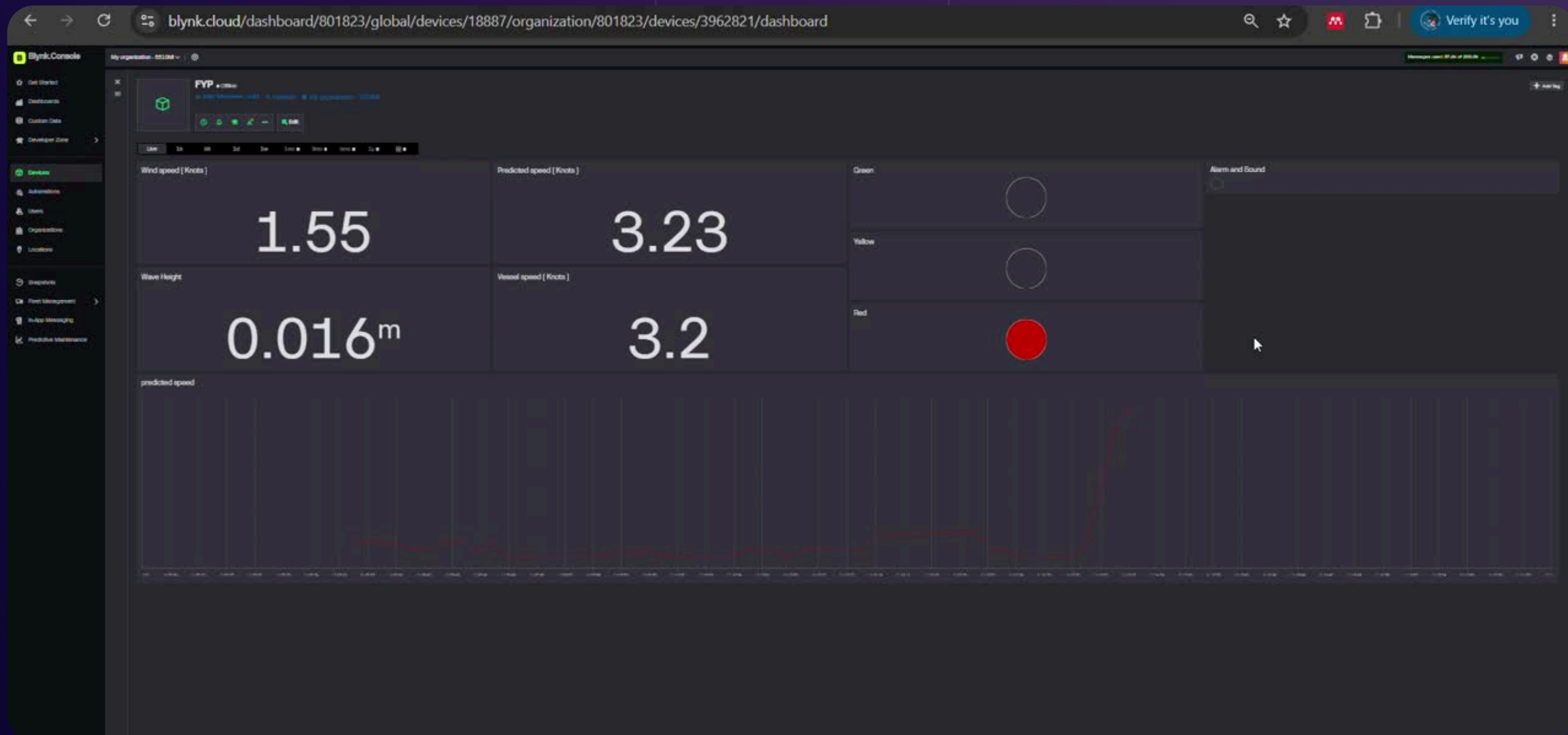
Phase	RMSE	R ²	MAPE
Offline	0.246	0.9991	—
Test 1	15.05	-2.99	65.40%
Test 2	6.59	-0.05	38.40%
Test 3	0.187	0.9989	2.09%

Three iterative road tests demonstrated progressive domain adaptation of the grey-box model, with each retraining cycle yielding measurable accuracy improvement

Road testing



Dashboard alerts



Conclusion

The project successfully achieved all objectives. The grey-box model demonstrated highly accurate vessel speed prediction with an RMSE of 0.074 knots and an R^2 of 0.9998, while 99.6% of predictions fell within ± 0.5 knots. The ESP32-based real-time monitoring system achieved 100% alert classification accuracy across all test sessions using the Blynk dashboard. Validation through three iterative road tests confirmed the reliability of the complete pipeline, from sensor data acquisition to prediction and alert generation. Future work will focus on deploying the system on maritime vessel for field validation and further model refinement using real-world at-sea data

Thank You